A Survey on Bayesian Network for AI

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***Abstract*—Bayesian networks (BNs) have evolved over three decades into a central formalism for probabilistic and causal reasoning in artificial intelligence, spanning structure learning theory, inference algorithms, and application-driven extensions. This survey synthesizes approximately 100 works published between 1993 and 2026, organized into seven thematic areas: causal discovery from observational and time-series data via dynamic and extended BN architectures; the role of probabilistic graphical models within broader AI systems such as decision-making, networking, and generative modeling; causal BNs tailored to medical diagnosis, prognosis, and clinical decision support; foundational algorithms for structure learning grounded in conditional independence testing; exact and approximate inference techniques for discrete, hybrid, and dynamic networks; metaheuristic and bio-inspired optimization strategies for structure search; and theoretical results characterizing the computational complexity of learning and inference. Across these themes, recurring trends include the fusion of BNs with deep learning and attention mechanisms for scalable causal discovery, growing emphasis on dynamic and streaming data settings, increasing adoption of hybrid exact-heuristic and metaheuristic search strategies to navigate NP-hard structure spaces, and persistent interest in explainability for high-stakes domains such as healthcare and policy. The review highlights unresolved challenges, including robust handling of latent confounders and feedback loops, principled benchmarking amid rapid proliferation of near-identical algorithmic variants, scalability of exact inference to high-dimensional and hybrid networks, and the need for standardized validation protocols. We conclude by outlining promising directions, including quantum-inspired inference, neuro-symbolic integration, and theoretically grounded approaches to bridging statistical causal discovery with interventional and counterfactual reasoning.**

***Index Terms*—Bayesian Networks, Probabilistic Graphical Models, Artificial Intelligence, Machine Learning, Causal Inference, Uncertainty Quantification, Knowledge Representation**

I. Introduction

BAYESIAN networks (BNs) have, for over three decades, served as one of the most principled frameworks for reasoning under uncertainty, combining the rigor of probability theory with the intuitive representational power of graphs [1]. By encoding a joint probability distribution as a directed acyclic graph of conditional dependencies, BNs allow practitioners to represent complex, high-dimensional systems compactly, to perform probabilistic inference in the presence of partial or missing evidence [2], [3], and, crucially, to support causal rather than purely correlational reasoning [4]. This dual capacity for probabilistic and causal inference has made Bayesian networks indispensable across a remarkable breadth of artificial intelligence applications, from medical diagnosis and prognosis [5]–[7] and neuroscience [8] to climate science [9], autonomous decision-making [10], and next-generation wireless network management [11]. As modern AI systems increasingly demand not only predictive accuracy but also transparency, uncertainty quantification, and causal understanding, Bayesian networks have re-emerged as a central tool for building trustworthy and explainable intelligent systems [12], [13].

The problem landscape surrounding Bayesian networks for AI is broad and multifaceted. At its core lie two intertwined challenges: structure learning, i.e., discovering the graphical skeleton and edge orientations that best represent the dependencies in observed data, and inference, i.e., computing posterior probabilities or causal queries given that structure. Structure learning itself spans constraint-based, score-based, and hybrid combinatorial algorithms [14], [15], exact dynamic-programming and integer-programming formulations that guarantee global optimality [16], [17], and a growing family of metaheuristic approaches—genetic algorithms, pigeon-inspired optimization, bacterial foraging, and opposition-based learning—that trade optimality guarantees for scalability [18]–[23]. Inference, in turn, ranges from exact algorithms exploiting causal independence and conditioning [24], [25] to approximate schemes such as loopy belief propagation, expectation propagation, and variational and neural-network-based approximations that scale inference to large or hybrid discrete-continuous networks [26]–[30]. Compounding these challenges is a long-standing theoretical result that both exact and approximate inference in general Bayesian networks are NP-hard [31], which has driven decades of work on tractable special cases, heuristics, and ordering strategies [32], [33]. Layered atop these foundations is the more recent and rapidly growing subfield of causal discovery from observational and streaming data using static and dynamic Bayesian networks [34]–[37], which seeks to recover not merely statistical association but the underlying causal mechanisms that generate data.

A dedicated survey is timely for several reasons. First, the field has recently undergone a marked resurgence, with a wave of new contributions on dynamic structure learning, latent-variable causal discovery, and hidden Markov model integration appearing within just the past year [38]–[42], reflecting renewed interest in causal AI and explainability. Second, Bayesian networks are increasingly being hybridized with deep learning—through neural parameterizations of graphical models, variational autoencoders with structured latent spaces, and Bayesian neural networks that approximate posterior uncertainty—blurring the boundary between classical probabilistic graphical models and modern representation learning [43]–[45]. Third, despite decades of algorithmic progress, foundational questions about computational tractability, scalability to high-dimensional and streaming data, and robustness to missing or imprecise data remain only partially resolved [31], [46], [47]. Existing surveys tend to focus narrowly on structure learning algorithms [15], [48] or on specific application domains such as neuroscience [8] or healthcare [5], leaving a gap for a unifying treatment that

connects causal discovery, inference algorithms, optimization-based learning, computational complexity, and cross-domain applications within a single coherent narrative centered on the role of Bayesian networks in contemporary AI.

The scope of this survey is accordingly broad but structured. We cover: (i) causal discovery methods that use static and dynamic Bayesian networks to recover causal structure from observational, streaming, and time-series data; (ii) the integration of probabilistic graphical models into broader AI systems, including hybrids with neural networks and generative models; (iii) causal Bayesian networks specifically applied to medical decision-making and diagnosis; (iv) the algorithmic and theoretical foundations of structure learning and conditional independence testing; (v) exact and approximate inference algorithms, including belief propagation, expectation propagation, and variational approaches; (vi) metaheuristic optimization techniques for structure learning; and (vii) the computational complexity results that bound what is achievable in both structure learning and inference. Throughout, we emphasize connections between classical probabilistic graphical model theory and its contemporary role in explainable, causal, and trustworthy AI.

The main contributions of this survey are as follows:

- We provide a unified taxonomy of Bayesian network research for AI, organizing the literature into causal discovery, broader probabilistic-graphical-model integration, medical decision-making applications, structure learning foundations, inference algorithms, metaheuristic optimization, and computational complexity.
- We synthesize recent advances in dynamic and streaming causal discovery, highlighting how latent-variable and hidden Markov model extensions address feedback loops and temporal dependencies [34], [37], [39].
- We critically compare exact and approximate inference paradigms, from classical belief propagation and expectation propagation to neural variational inference, clarifying their trade-offs in accuracy, scalability, and applicability to hybrid networks [26]–[29].
- We consolidate structure-learning algorithms spanning constraint-based, score-based, exact, and metaheuristic optimization methods, discussing their comparative strengths and the known complexity barriers that constrain them [15], [16], [19], [31].
- We examine the role of Bayesian networks in explainability and medical AI, illustrating how causal structure supports interpretable diagnostic and decision-support systems [6], [12], [13].
- We identify open challenges and promising research directions at the intersection of Bayesian networks, deep learning, and causal AI, intended to guide future work in this space.

The remainder of this survey is organized as follows. Section II describes the methodology used to collect and organize the literature reviewed in this survey. Section III introduces the background and theoretical foundations of Bayesian networks, including probabilistic semantics, conditional independence, and causal interpretation. The subsequent thematic sections present, in turn, causal discovery methods, the integration of probabilistic graphical models into broader AI systems, causal Bayesian networks for medical decision-making, structure learning and conditional independence foundations, exact and approximate inference algorithms, metaheuristic optimization for structure learning, and the computational complexity of these tasks. Section XI discusses cross-cutting trends and synthesizes insights across these themes. Section XII outlines open challenges and promising directions for future research. Finally, Section XIII concludes the survey.

TABLE I
SEARCH AND SELECTION STATISTICS

Stage	**Records**
OpenAlex	420
Semantic Scholar	60
Total retrieved	480
After de-duplication	443
After open-access/date filtering	426
Included in survey	100
journal articles	51
conference papers	1
preprints	48
other	0

II. SURVEY METHODOLOGY

The literature was collected on October 13, 2026 from OpenAlex, Semantic Scholar, which index openly accessible scholarly works. The following search queries were issued to every source:

- `Bayesian Network for AI`
- `Bayesian network structure learning`
- `probabilistic graphical models inference`
- `causal discovery Bayesian networks`
- `dynamic Bayesian networks AI`
- `Bayesian networks medical diagnosis`
- `approximate inference belief networks`

Records of the same work returned by different sources or queries were merged using DOI, arXiv identifier and normalised title; when a preprint had a published version, the version of record was retained for citation. Only works with an open-access full text were kept. The remaining works were ranked by a weighted score combining TF-IDF relevance to the topic and queries (70%), log-scaled citation count (20%) and recency (10%); works without abstracts or with negligible relevance were discarded, and at most the 100 highest-scoring works were retained. Selected works were grouped into themes by k-means clustering of their TF-IDF representations. Table I summarises the selection process.

III. BACKGROUND

A. Core Concepts and Terminology

A Bayesian Network (BN) is a probabilistic graphical model that represents a joint probability distribution over a set of random variables $X = \{X_1, X_2, \ldots, X_n\}$ by combining a

qualitative graphical component with a quantitative probabilistic component [8], [48]. The graphical component is a directed acyclic graph (DAG) $G = (V, E)$, where each node $v_i \in V$ corresponds to a random variable X_i and each directed edge $(X_j \to X_i) \in E$ encodes a direct probabilistic dependency of X_i on X_j, often interpreted, under additional causal assumptions, as X_j being a direct cause of X_i [15]. The quantitative component associates with every node X_i a conditional probability distribution $P(X_i \mid \mathrm{Pa}(X_i))$, where $\mathrm{Pa}(X_i)$ denotes the parents of X_i in G. The defining property of a BN is the local Markov assumption, which states that each variable is conditionally independent of its non-descendants given its parents; this assumption licenses the chain-rule factorisation of the joint distribution as

$$P(X_1, X_2, \ldots, X_n) = \prod_i = 1\hat{}nP\big(X_i \mid \mathrm{Pa}(X_i)\big). \tag{1}$$

This factorisation is what makes BNs computationally and cognitively tractable relative to a naively specified joint distribution, since each local conditional distribution depends only on a (typically small) parent set rather than on all other variables. Depending on the nature of the X_i, conditional distributions may be represented as conditional probability tables (discrete variables), conditional Gaussians, or more general parametric or nonparametric forms in hybrid networks that mix discrete and continuous variables [29]. Extensions of the basic formalism include dynamic Bayesian networks, which unroll the DAG over discrete time slices to model temporal processes, and causal Bayesian networks, which augment the graph with interventional semantics so that the effect of a do-operation can be distinguished from passive conditioning [8], [9].

B. Standard Problem Formulation

Work on BNs is conventionally organised around three canonical tasks: structure learning, parameter learning, and inference. Structure learning seeks to recover a DAG G (or an equivalence class thereof) from a dataset $D = \{x^{(1)}, \ldots, x^{(N)}\}$ of observed instantiations of X. Two broad methodological families are used: constraint-based methods, which test conditional independence relations among variables and orient edges according to resulting independence patterns, and score-based methods, which search the space of DAGs to optimise a scoring function $\mathrm{Score}(G, D)$ (e.g., BIC or Bayesian Dirichlet scores) that trades off data fit against model complexity, as well as hybrid approaches that combine both ideas [14], [15]. Because the number of possible DAGs grows super-exponentially with n, exact structure learning is itself an NP-hard combinatorial optimisation problem, motivating heuristic search, local learning, and restriction to sparse or bounded-treewidth structures [14], [15]. Parameter learning, given a fixed structure G, fits the local conditional distributions $P(X_i \mid \mathrm{Pa}(X_i); \theta_i)$ to data, typically via maximum likelihood or Bayesian estimation with conjugate priors over $\theta = (\theta_1, \ldots, \theta_n)$.

Inference is the task of computing posterior marginal or joint probabilities of a subset of query variables Q given observed evidence $E = e$,

$$P(Q \mid E = e) = \frac{P(Q, E = e)}{P(E = e)} = \frac{\sum_ZP(Q, E = e, Z)}{\sum_Q, ZP(Q, E = e, Z)}, \tag{2}$$

where Z denotes the remaining, unobserved variables. Exact inference can be carried out efficiently in tree-structured or low-treewidth networks via message-passing algorithms such as belief propagation and junction-tree (clique-tree) methods that exploit factorisation and, where present, causal-independence structure to reduce computational cost [24]. In general, however, exact inference in arbitrary BNs is NP-hard, and even approximating the posterior probabilities to within guaranteed bounds remains NP-hard [31]. This intractability motivates approximate inference schemes, including loopy belief propagation on networks with cycles [26], expectation propagation for hybrid discrete-continuous networks [27], and sampling-based or variational approaches more broadly [29].

C. Positioning Relative to Existing Surveys

Several prior surveys have charted parts of this landscape but typically with a narrower methodological or application-specific scope. Kitson et al. provide an extensive and up-to-date survey of combinatoric BN structure-learning algorithms, cataloguing 74 methods but deliberately excluding inference and application-oriented discussion [15]. Bielza and Larrañaga review both structure learning and inference algorithms but frame their survey specifically around applications in neuroscience [8]. Salmerón et al. focus narrowly on inference algorithms for hybrid networks that mix discrete and continuous variables [29], while Zhou offers a broader but considerably older treatment of structure learning across probabilistic graphical models in general, rather than BNs specifically [48]. A recent narrative review traces the historical arc of probabilistic graphical models from Pearl's early inference engines through causal reasoning and modern machine learning references, but it is explicitly historiographic rather than a technical synthesis of current methods [49]. Domain-specific reviews, such as those surveying BNs for disease diagnosis and prognosis [5] or cataloguing their distribution across medical conditions [50], further illustrate the breadth of BN adoption but are confined to healthcare applications and do not engage with the underlying structure-learning or inference methodology in depth. In contrast to these efforts, the present survey aims to integrate the methodological core of BNs—representation, structure learning, parameter learning, and both exact and approximate inference—with their expanding role within modern AI systems, including hybridisation with neural architectures [43], [44], causal discovery [9], [51], and deployment across diverse domains such as phylogenetics [52] and wireless network management [11], thereby providing a unified treatment that bridges foundational theory and contemporary application that is not fully covered by any single prior review.

IV. Bayesian Networks for Causal Discovery from Data

Causal discovery seeks to recover the directed dependency structure that generates observational (and sometimes interventional) data, and Bayesian networks have long served as the

TABLE II
SUMMARY OF WORKS ON BAYESIAN NETWORKS FOR CAUSAL DISCOVERY FROM DATA (PART 1)

Ref.	Year	Venue	Main contribution
[53]	2017	Journal of Artificial Intelligence Res...	Activator dynamic Bayesian networks model indirect causal influences over time
[51]	2019	Machine Learning and Knowledge Extract...	Attention-based convolutional neural network framework discovers causal graphs from time series
[54]	2020	IEEE Access	Models topic-sentiment dynamics using Gaussian Process Dynamic Bayesian Networks
[34]	2026	arXiv	Extends Bayesian network structure learning to dynamically adapt to evolving causal relationships in time-series data
[35]	2026	arXiv	Uses dynamic Bayesian networks with variational inference to continuously adapt structure for streaming observational data
[38]	2026	arXiv	Develops a Bayesian inference framework for dynamic Bayesian networks to mitigate confounding and model temporal dependencies
[7]	2026	arXiv	Applies Causal Bayesian Networks to model symptom-disease-treatment relationships for improved medical diagnosis
[12]	2026	arXiv	Uses Causal Bayesian Networks to generate causal explanations for AI-based medical diagnostic predictions
[39]	2026	arXiv	Integrates latent variables via Hidden Markov Models into dynamic Bayesian networks to handle feedback loops
[40]	2026	arXiv	Combines Bayesian networks with Hidden Markov Models for causal discovery in time-series
[37]	2026	arXiv	Dynamic Bayesian Networks with new MCMC method infer causal structure from observational data
[55]	2026	arXiv	Learns dynamic Bayesian networks via PC algorithm to capture temporal causal relationships
[56]	2026	arXiv	Introduces Bayesian Copula Networks for causal discovery under complex dependencies
[57]	2026	arXiv	Proposes data-driven Bayesian Network-Based Causal Discovery Algorithm for inferring causal graphs

TABLE III
SUMMARY OF WORKS ON BAYESIAN NETWORKS FOR CAUSAL DISCOVERY FROM DATA (PART 2)

Ref.	Year	Venue	Main contribution
[58]	2026	arXiv	Develops Bayesian network structure learning algorithm for high-dimensional temporal causal discovery
[59]	2026	arXiv	Quantum-inspired probabilistic amplitudes enable parallel exploration of causal structures
[60]	2026	arXiv	Intervention-aware Bayesian network extension for causal discovery from streaming data
[61]	2026	arXiv	Combines Temporal Bayesian Networks with causal discovery to model causal dynamic dependencies
[62]	2026	arXiv	Integrates Granger causality tests within Dynamic Bayesian Networks for improved causal inference
[63]	2026	arXiv	Combines neural feature extraction with Dynamic Bayesian Networks for adaptive medical diagnosis
[64]	2026	arXiv	Augments adaptive Bayesian network learning with symbolic reasoning for interpretable causal discovery
[65]	2026	arXiv	Uses deep neural networks to learn Bayesian network structure for causal discovery
[66]	2026	arXiv	Integrates graph-based reasoning with Bayesian networks for iterative causal structure discovery
[67]	2026	arXiv	Jointly learns Bayesian network structure and parameters via variational inference for causal inference
[68]	2026	arXiv	Combines episodic memory networks with Bayesian inference for causal discovery
[69]	2026	arXiv	Combines Granger causality tests with Bayesian networks for real-time streaming causal discovery
[70]	2026	arXiv	Proposes iterative graph-based Bayesian network algorithm refining causal graphs via conditional independence tests
[71]	2026	arXiv	Develops dynamic Bayesian networks with time-varying Markov chain adapting structure for explainable time series

TABLE IV
SUMMARY OF WORKS ON BAYESIAN NETWORKS FOR CAUSAL DISCOVERY FROM DATA (PART 3)

Ref.	Year	Venue	Main contribution
[72]	2026	arXiv	Combines Bayesian networks with Granger causality for streaming causal discovery

canonical representation for this task because their directed acyclic graphs (DAGs) can be interpreted, under assumptions such as the causal Markov condition and faithfulness, as encoding genuine causal mechanisms rather than mere statistical association. Classical structure-learning algorithms for static BNs are well understood, but two persistent difficulties have driven much of the recent literature: first, real systems frequently exhibit cyclic, feedback-laden, or time-lagged dependencies that a single static DAG cannot represent; second, learning structure directly from high-dimensional, noisy, and non-stationary observational data without strong prior knowledge remains computationally and statistically hard. The works reviewed below can be organized around how they address these two difficulties — through explicit temporal/dynamic modeling, through hybridization with other inference paradigms, through alternative structural representa-

tions and priors, and through domain-specific applications — while also revealing a growing split between methods validated on real data and a large body of recent, purely theoretical proposals.

A. Temporal and Dynamic Bayesian Networks for Causal Discovery

An early and influential line of work recognized that naively discretizing time in dynamic Bayesian networks (DBNs) can produce spurious causal conclusions when indirect influences are not anticipated correctly. [53] show that modeling causal dependencies that require cycles at a coarse temporal granularity, without a "causal design" for how indirect effects unfold over time, leads standard DBNs astray; their activator dynamic Bayesian networks (ADBNs) introduce activator random variables that let the network rapidly adapt to context while preserving classical BN semantics and inference algorithms, without additional modeling or computational overhead. This result is foundational because it demonstrates, with mathematical rigor, that the choice of temporal resolution and node semantics is itself a causal modeling decision, not a mere implementation detail — a concern echoed, with less formal grounding, by much of the later literature on dynamic structure learning.

Moving from classical graphical-model semantics toward deep learning, [51] propose the Temporal Causal Discovery Framework (TCDF), which uses attention-based convolutional neural networks together with a causal validation step to learn temporal causal graphs, including time delays, confounders, and instantaneous effects, directly from continuous time-series data. Evaluated on financial and neuroscientific benchmarks, TCDF reports state-of-the-art performance and shows that interpreting the internal attention weights of a trained network can circumstantially reveal hidden confounders — illustrating a complementary strategy to the explicit activator-variable design of [53]: rather than hand-crafting the causal semantics of time, TCDF lets a flexible function approximator discover lagged dependencies, at the cost of the transparent probabilistic semantics that classical BNs provide. A third empirically grounded contribution, [54], applies a Gaussian Process Dynamic Bayesian Network to model how topic sentiment evolves and interacts over time, learning relationships between the sentiment trajectories of related topics on a large real-world Twitter corpus (9.72 million tweets) and presenting a case study on Brexit-related discourse. Together these three works mark the few entries in this survey subset with genuine empirical evaluation on real or benchmark data, in contrast to the large cluster discussed next.

A substantial portion of the recent literature consists of proposals that extend static BN structure-learning algorithms to time-varying settings but remain largely conceptual. [34], [35], and [37] each argue that traditional BN structure learning is ill-suited to evolving causal relationships and each proposes a variant of dynamic structure learning — respectively, adapting structure-learning algorithms to temporal shifts, using variational inference to continuously update topology and parameters as data streams arrive, and introducing an MCMC scheme for inference within an evolving DBN structure. [55] similarly proposes building DBNs with PC-style structure learning while explicitly incorporating time-varying covariates and feedback effects, [61] integrates a causal discovery algorithm on top of temporal Bayesian networks to recover a DAG over time-indexed variables, and [58] targets high-dimensional time series specifically, proposing temporal Markov assumptions combined with efficient inference to scale dynamic structure learning. These papers converge on a shared diagnosis — that non-stationarity and feedback undermine static conditional-independence-based discovery — and a shared remedy of iteratively or adaptively updating BN topology, but, unlike [53], [51], and [54], none report empirical results on concrete datasets, presenting instead theoretical frameworks, proposed methodologies, or conceptual outlines; this leaves open whether the proposed dynamic-adaptation mechanisms yield measurable gains over established DBN learning or deep temporal causal discovery approaches.

B. Hybrid and Deep Architectures for Causal Discovery

A second cluster of work seeks to enrich BN-based causal discovery by hybridizing it with complementary temporal or statistical machinery. [40] and [39] both integrate Hidden Markov Models into the BN learning pipeline to capture latent, time-varying dependencies; the latter further targets feedback loops explicitly, using variational and approximate inference to manage the resulting computational burden of jointly representing latent states and cyclic structure. These proposals extend the latent-variable handling that classical BN learning struggles with, but, as with the dynamic-structure cluster above, their claims of improved robustness to latent confounding and cycles are argued theoretically and conceptually rather than demonstrated on data.

A closely related set of papers revisits Granger causality — a long-standing, purely predictive notion of temporal causation — and proposes embedding it within a Bayesian network to add graphical, probabilistic structure to Granger-style tests. [62] constructs a DBN to represent temporal dependencies and then runs Granger tests within that network to mitigate bias and confounding; [69] and [72] apply essentially the same combination of Granger causality and Bayesian networks to the streaming-data setting, iteratively updating network structure as Granger-causality evidence accrues in real time. The near-duplication across these three proposals — all pairing Granger tests with BN structure refinement, two of them specifically for streaming scenarios — suggests an active but not yet consolidated line of inquiry into reconciling predictive Granger causality with the graphical semantics of Bayesian networks, again without reported empirical comparison to either pure Granger methods or standard DBN learning.

Other hybridizations explore alternative computational substrates for structure search. [65] proposes training a deep neural network to learn conditional independence structure directly from data and extracting a Bayesian network from the trained model, aiming to reduce reliance on strong prior assumptions in the spirit of the attention-based approach of [51], but evaluated only conceptually. [59] takes a more speculative

route, proposing to represent and explore multiple candidate causal structures concurrently via probabilistic amplitudes inspired by quantum superposition, intended to address the combinatorial explosion of candidate DAGs that limits classical search-based structure learning; this idea is presented purely at the level of theoretical motivation. Finally, two works connect dynamic BN causal discovery to applied, intervention-rich settings: [63] couples a feature-extracting neural network with a dynamically updating DBN for diagnostic reasoning, and [60] extends BNs with explicit intervention models to correct the bias that naive causal discovery incurs when interventions are occurring within a streaming data source. Both target practically important scenarios — diagnosis and intervention-contaminated streams — but, like most of this cluster, stop at a formalized framework rather than empirical validation.

C. Structural Representations, Priors, and Interpretability

A further strand of work focuses less on temporal dynamics and more on the representational and algorithmic machinery used to search over DAG space. [66] and [70] both propose graph-based Bayesian networks that iteratively refine a candidate causal graph using PC-algorithm-style conditional independence tests, framed as a scalable alternative for high-dimensional data; the two papers are substantively similar in motivation and mechanism, differing mainly in emphasis on general graph reasoning versus explicit scalability claims for observational data, and both rely on theoretical argument rather than experiments to support their scalability and accuracy claims.

Complementary efforts aim to enrich the probabilistic representation underlying BN causal discovery. [56] introduces Bayesian Copula Networks, which encode variable dependencies via copulas before applying Bayesian inference to estimate causal relationships, targeting the intricate dependency, confounding, and selection-bias patterns that standard conditional-independence tests can miss. [67] instead addresses the joint learning of graph structure and parameters simultaneously via variational inference, incorporating prior knowledge to improve robustness relative to manually specified structures. [57] proposes a more baseline-oriented Bayesian Network-Based Causal Discovery Algorithm (BNCDA) that learns conditional independence relationships from data to infer a causal graph without heavy reliance on domain expertise, positioning itself as a purely data-driven entry point into this family. Across these three papers the recurring theme is a search for richer statistical representations (copulas, joint structure-parameter posteriors, direct CI-based graph inference) to loosen the dependence of BN causal discovery on strong prior specification, though again each is presented at the level of theoretical framework and conceptual demonstration.

A smaller set of papers emphasizes interpretability and reasoning under partial observability. [64] augments Bayesian network learning with a symbolic reasoning engine that injects domain rules and logical constraints, arguing that purely data-driven discovery is prone to spurious correlations and poor interpretability, and that adaptively combining statistical evidence with symbolic knowledge yields more robust, explainable causal models. [68] instead draws on episodic memory networks, representing states and causal transitions as nodes and edges and applying Bayesian inference to estimate transition probabilities, with the stated aim of supporting causal discovery in partially observable environments by simulating interventions over remembered state trajectories. Both works target known weaknesses of vanilla statistical structure learning — spurious edges and incomplete observability — through structurally distinct but conceptually aligned uses of auxiliary memory or knowledge representations, again without empirical benchmarking.

D. Applications to Medical Diagnosis and Explainable AI

A final cluster applies causal Bayesian networks specifically to domains where interpretability and causal (rather than merely correlational) reasoning are explicitly valued. [7] constructs a causal Bayesian network linking symptoms, diseases, and treatments for medical diagnosis, arguing that explicit causal modeling better supports reasoning about interventions than standard, association-based Bayesian networks, particularly under confounding. [12] extends this motivation directly into explainable AI, proposing to generate diagnostic explanations grounded in the causal pathways of a CBN rather than in purely correlational model outputs, aiming to increase clinical trust and transparency. [71] generalizes the explainability goal to general time-series settings, proposing a DBN whose structure evolves via a time-varying Markov chain so that the resulting explanations reflect how temporal patterns shape variable relationships. These three works share a normative argument — that causal, graphical structure is preferable to black-box correlational prediction for high-stakes, interpretability-sensitive applications — but, consistent with the rest of the 2026 cluster discussed throughout this section, their support for this argument remains largely illustrative and conceptual rather than empirically demonstrated on clinical or benchmark time-series data.

Taken together, the literature on Bayesian-network-based causal discovery shows a clear methodological trajectory: from carefully justified, mathematically grounded treatments of cyclic and temporal causal semantics in classical DBNs [53], through empirically validated deep-learning-based temporal causal discovery [51] and applied dynamic BN modeling of real social-media data [54], to a much larger, recent body of proposals that extend BN/DBN structure learning with dynamic adaptation, latent-variable and HMM integration, Granger-causality hybridization, copula representations, symbolic and episodic-memory reasoning, intervention-awareness, and medical/explainability applications [34], [35].

V. Probabilistic Graphical Models in Broader AI Systems

Probabilistic graphical models (PGMs) rarely operate today as isolated, stand-alone systems; increasingly they appear as interpretable or uncertainty-aware components embedded within larger artificial intelligence pipelines spanning deep learning, networked infrastructure, scientific computing, and applied decision support. This section surveys that broader

TABLE V
SUMMARY OF WORKS ON PROBABILISTIC GRAPHICAL MODELS IN BROADER AI SYSTEMS (PART 1)

Ref.	Year	Venue	Main contribution
[2]	2013	arXiv	Uses causal Bayesian networks to enable consistent inference despite missing not-at-random data
[43]	2016	arXiv	Combines graphical models with neural network observation models for structured representation learning and fast variational inference
[52]	2016	Systematic Biology	RevBayes software enables flexible graphical-model-based Bayesian phylogenetic inference
[11]	2020	IEEE Open Journal of Vehicular Technol...	Proposes AI-integrated network-slicing architecture for next-generation wireless networks
[73]	2020	Annual Meeting of the Association for ...	Stacks Bayesian network ensembles on entity-aware CNNs for interpretable clinical diagnosis
[74]	2020	Big Data Mining and Analytics	Proposes inference attacks predicting unknown genomic SNPs/traits using probabilistic graphical models and belief propagation
[75]	2021	IEEE Transactions on Information Theor...	Connects multi-marginal optimal transport to Bayesian inference in graphical models
[76]	2021	Scientific Reports	DropConnect approximates Bayesian inference to model deep network uncertainty
[45]	2023	Entropy	Uses generative deep learning mappings to reconstruct posterior distributions for density-free Bayesian computation
[6]	2024	International Journal of Intelligent S...	Builds an AI doctor assistant using adaptive Bayesian networks for diagnosis
[5]	2024	Machine Learning and Knowledge Extract...	Scoping review of Bayesian networks applied to disease diagnosis and prognosis
[77]	2024	Internet of Things	Fishnet-based AI and Bayesian game-theoretic resource allocation for 6G IoT networks
[10]	2025	Symmetry	Integrates probabilistic graphical models, MCMC, and Bayesian optimization for autonomous vehicle navigation decision-making
[78]	2025	Sustainability	Develops a neural network trained via Bayesian regularization to predict larval protein conversion yield

TABLE VI
SUMMARY OF WORKS ON PROBABILISTIC GRAPHICAL MODELS IN BROADER AI SYSTEMS (PART 2)

Ref.	Year	Venue	Main contribution
[79]	2025	Electronics	Uses Bayesian network-based structural equation model to explain generative AI use intention

integration, tracing how Bayesian-network and PGM machinery has been combined with neural architectures, deployed in clinical and diagnostic AI, folded into networked and autonomous cyber-physical systems, and used as a foundational inference substrate for domains as varied as phylogenetics, optimal transport, and privacy analysis. Across these threads, a common pattern emerges: PGMs supply structure, interpretability, and principled uncertainty quantification, while neural networks, metaheuristics, or domain-specific pipelines supply scalability and representational power.

A. Hybrid Neural–Graphical Architectures for Representation Learning and Uncertainty

A first line of work seeks to fuse the structural and inferential advantages of graphical models with the representational capacity of deep networks. [43] compose explicit graphical-model latent structure with neural observation models, extending variational autoencoders so that recognition networks output conjugate potentials consumable by graphical-model message passing; the resulting structured variational autoencoder is trained end-to-end with stochastic variational inference, natural gradients, and the reparameterization trick, and is demonstrated on behavioral phenotyping. In contrast, [76] pursue Bayesian-style uncertainty without retaining an explicit graph: Monte Carlo DropConnect approximates Bayesian inference in deep networks by imposing a Bernoulli distribution over weights, yielding calibrated uncertainty estimates for classification and segmentation at a fraction of the computational cost of traditional Bayesian neural networks. [45] push the hybridization further by inverting the usual role of generative AI, using deep quantile networks as a density-free mechanism to directly map simulated data to posterior draws, positioning this "BayesGen-AI" framework as an alternative to MCMC, ABC, and vanilla GANs for Bayesian computation itself.

These three approaches illustrate a spectrum of design choices rather than a single consensus method. [43] preserves conjugate graphical structure explicitly, trading some representational flexibility for interpretable, efficient message-passing inference; [76] and [45] instead approximate Bayesian behavior purely through network training procedures (dropout-like regularization, deep quantile regression), sacrificing an explicit dependency graph for scalability to large architectures and avoidance of intractable density estimation. The shared limitation across this subsection is that fidelity to the true posterior is only loosely guaranteed—message-passing approximations in [43] depend on conjugacy assumptions that may not hold broadly, while the regularization-based uncertainty of [76] and the architecture choices required in [45] (e.g., feature selection and nonlinear design) introduce their own sources of approximation error that are difficult to quantify theoretically.

B. Probabilistic Graphical Models in Clinical and Diagnostic AI

Healthcare is one of the most active application areas for embedding PGMs inside larger diagnostic systems, driven by the field's simultaneous need for predictive accuracy and interpretability. [73] stack Bayesian network ensembles on top of entity-aware convolutional neural networks for text-based clinical diagnosis, using the CNN component to extract accurate entity-aware representations from electronic medical records while the Bayesian network layer provides a human-readable causal explanation of the diagnosis; evaluated on real, doctor-annotated EMR data, the framework reportedly outperforms prior automatic diagnosis methods on both accuracy and interpretability criteria. [6] propose a related but less rigorously validated system—an "AI doctor assistant" that constructs a Bayesian network over patient history, symptoms, and diagnostic factors, continuously updated as new data arrive—though the work provides limited quantitative benchmarking of its diagnostic performance compared to [73]. Complementing these individual systems, the scoping review by [5] synthesizes 60 studies applying Bayesian networks to disease diagnosis and prognosis across diverse medical domains, reporting an average AUC exceeding 75% and documenting the method's broad applicability and feasibility of implementation.

Read together, these works reveal a maturing but uneven field: [73] exemplifies a principled hybridization in which the Bayesian network component is tightly coupled to a learned neural encoder with explicit evaluation of interpretability, whereas [6] reflects a more exploratory, less benchmarked application of the same idea. The review by [5] provides crucial context, confirming strong average predictive performance across the literature but also implicitly highlighting inconsistencies in validation rigor and reporting standards that individual application papers such as [6] exemplify—underscoring an open need for standardized evaluation protocols when PGMs are deployed as diagnostic decision-support layers.

C. PGMs in Networked, Autonomous, and Cyber-Physical Systems

Beyond healthcare, probabilistic and Bayesian reasoning have been proposed as components of AI-driven management for networked and autonomous infrastructure

VI. Causal Bayesian Networks for Medical Decision-Making

Causal reasoning is central to medical decision-making: clinicians do not merely want to predict a diagnosis or outcome but to understand which interventions, risk factors, or evidence actually drive it. Bayesian networks (BNs), by encoding conditional dependencies as a directed acyclic graph with associated conditional probability tables, offer a natural substrate for this kind of reasoning because they can be read either as a compact factorization of a joint probability distribution or, under additional causal assumptions, as a causal model supporting interventional and counterfactual queries. This dual interpretation has made BNs attractive in clinical and policy domains where transparency, auditability, and the ability to reason from cause to effect (and vice versa) are as important as raw predictive accuracy. The literature surveyed here spans foundational overviews of BN adoption in medicine, temporally-aware diagnostic models, a cluster of recent work on causal discovery and causal decision-making, and extensions of BN-based causal reasoning to non-clinical, but structurally similar, decision domains such as policy and law.

A. Foundational Surveys and the Case for Bayesian Networks in Healthcare

A first strand of work consolidates evidence for why BNs are well suited to medicine and maps out where they have actually been deployed. [8] survey the use of BNs in neuroscience, covering structure learning and exact/approximate inference, and observe that despite their generality for handling discrete, continuous, and temporal variables, BNs remain comparatively under-used relative to their potential, with applications concentrated on functional connectivity analysis and classification from neuroimaging and omics data. Complementing this, [50] conduct a broader meta-analysis of BN applications across medical conditions and find that nearly two-thirds of published healthcare BNs cluster around just four condition types—cardiac disease, cancer, psychological disorders, and lung disease—suggesting that adoption has been opportunistic rather than systematic, and that a persistent gap in clinician understanding of BN capabilities limits translation into routine practice. [82] narrow the focus to radiology, cataloguing diagnostic and treatment-planning applications and highlighting a key distinguishing feature of BNs relative to deep learning: the ability to integrate observational findings with canonical domain knowledge, reason bidirectionally (cause-to-effect and effect-to-cause), and explain inferences, while noting that BNs have not themselves been used for computer vision and are more commonly combined with deep learning in hybrid pipelines where a neural model extracts image findings and a BN performs the probabilistic diagnostic synthesis.

This emphasis on interpretability is sharpened by contrast with purely connectionist approaches. [80] review artificial neural networks in medical diagnosis and, while documenting their capacity to integrate heterogeneous clinical, biochemical, and imaging data, implicitly underscore the "black-box" limitation that motivates graphical, probabilistic alternatives such as BNs. Pedagogical and introductory treatments, such as [1], reinforce this positioning by framing Bayes' rule and BNs as foundational tools for probabilistic reasoning under uncertainty in AI more broadly, though such accounts remain largely expository rather than empirical. Taken together, this group of papers establishes a consistent narrative: BNs are repeatedly praised for transparency and the ability to fuse evidence with prior knowledge, yet their clinical footprint remains narrow and uneven, concentrated in a handful of high-burden conditions and rarely benchmarked against modern deep learning baselines on shared datasets.

TABLE VII
SUMMARY OF WORKS ON CAUSAL BAYESIAN NETWORKS FOR MEDICAL DECISION-MAKING (PART 1)

Ref.	Year	Venue	Main contribution
[80]	2013	Journal of Applied Biomedicine	Reviews artificial neural network capabilities and limitations for medical diagnosis applications
[81]	2013	arXiv	Proposes Temporal Nodes Bayesian Networks representing multi-granularity temporal events for diagnosis and prediction
[8]	2014	Frontiers in Computational Neuroscienc...	Survey of Bayesian network structure learning and inference applications in neuroscience
[50]	2020	Artificial Intelligence in Medicine	Surveys distribution of Bayesian network applications across different medical conditions in healthcare
[1]	2023	INTERANTIONAL JOURNAL OF SCIENTIFIC RE...	Surveys foundational concepts and applications of Bayes' Rule and Bayesian networks in AI
[82]	2023	Radiology Artificial Intelligence	Reviews Bayesian network applications and advantages in radiology diagnosis and planning
[83]	2023	Cureus	Reframes medical diagnosis as Bayesian reasoning and probabilistic elimination process
[84]	2025	Data & Policy	Applies Bayesian causal structure learning to inform policy decision-making pathways
[85]	2025	Systems	Uses Bayesian networks to support transparent, evidence-based judicial decision-making
[4]	2026	Entropy	Reviews Bayesian networks and causal discovery methods for representing complex data relationships
[86]	2026	arXiv	Bayesian network framework with causal inference for improved medical diagnosis accuracy
[41]	2026	arXiv	Framework combining Bayesian network structure learning with decision theory for optimal decisions
[36]	2026	arXiv	Dynamic Bayesian Networks incrementally learn causal structure from streaming data
[42]	2026	arXiv	Probabilistic graphical model framework learns causal structure to answer counterfactual queries

TABLE VIII
SUMMARY OF WORKS ON CAUSAL BAYESIAN NETWORKS FOR MEDICAL DECISION-MAKING (PART 2)

Ref.	Year	Venue	Main contribution
[87]	2026	arXiv	Uses intervention simulations within Bayesian networks to mitigate confounding in causal discovery

B. Temporal and Diagnostic Reasoning Models

A second line of work operationalizes BNs specifically for the sequential, evidence-updating nature of diagnosis. [81] address a core limitation of standard BNs—their lack of native temporal semantics—by proposing Temporal Nodes Bayesian Networks (TNBN), in which each node represents an event or state change and arcs encode causal-temporal relationships with variable-granularity intervals. Benchmarked against a dynamic Bayesian network on a medical toy example and evaluated empirically on a fossil power plant fault-diagnosis task, TNBNs are reported to achieve good diagnostic and predictive performance while being more compact than fully unrolled dynamic BNs, illustrating a recurring trade-off in this literature between temporal expressiveness and model/inference complexity.

Other work targets the conceptual alignment between BN inference and the clinical reasoning process itself rather than introducing new structural formalisms. [83] argue, through a case study of Kartagener syndrome, that differential diagnosis is intrinsically Bayesian belief updating, and use this framing to demystify AI-assisted diagnosis for practitioners, positioning Bayesian reasoning as an interpretable complement to—rather than replacement for—clinical judgment; the contribution is conceptual and pedagogical rather than a deployed system. [86] revisit this problem computationally, proposing a framework in which a BN explicitly encodes causal links between diseases and symptoms and is combined with causal inference algorithms to update diagnostic probabilities under confounding and uncertainty. Relative to [81], which emphasizes temporal granularity, and [83], which emphasizes conceptual correspondence, [86] emphasizes causal robustness, but like several other recent entries in this literature it is presented without empirical validation on real clinical datasets, leaving open how these causal-diagnostic frameworks perform against established BN or machine-learning baselines in practice.

C. Causal Discovery, Structure Learning, and Decision-Theoretic Extensions

A third, more methodologically oriented cluster shifts attention from fixed, expert-elicited BN structures toward learning causal structure directly from data and coupling it with decision-making. [4] frame causal discovery as foundational to AI research generally, situating BN structure learning within the broader causal discovery literature. Building on this premise, a series of works by [41], [36], [42], and [87] each propose variations on the theme of learning causal BN structure and using it for inference: [41] combines structure learning with decision theory to select optimal actions from a learned causal model; [36] extends this to streaming data via Dynamic Bayesian Networks that incrementally update structure and parameters as the underlying distribution evolves, targeting settings where complete datasets are never simultaneously available; [42] applies probabilistic graphical models to counterfactual query answering, explicitly aiming to

disentangle correlation from causation; and [87] proposes incorporating simulated interventions into the structure-learning objective itself as a way of mitigating confounding bias that plagues purely observational causal discovery.

These four contributions share both strengths and a common limitation. Methodologically, they converge on a similar pipeline—structure learning, followed by causal/counterfactual inference, followed in some cases by decision optimization—and each targets a genuine open problem in applying BNs causally (streaming data, counterfactuals, confounding, decision integration). However, by the authors' own framing, several of these works are explicitly theoretical or conceptual, lacking empirical validation on real datasets, which makes it difficult to assess whether the proposed mechanisms (e.g., intervention simulation in [87], or incremental DBN updating in [36]) yield practical gains over established structure-learning algorithms such as those surveyed in [8] or [4]. In this sense they read as a research agenda for causal BN methodology rather than as validated clinical tools. By contrast, [84] ground the same structure-learning philosophy in an applied setting, learning BN structure to differentiate proximal from upstream causal factors and explicitly retaining multiple near-equally-probable causal pathways rather than committing to a single most-likely graph; applied to Australian educational policy data (e.g., student attendance and social disadvantage), this work demonstrates that the probabilistic, multi-pathway view of causal discovery can yield actionable, interpretable insight for policy decision-making, offering a template that medical causal BN research could emulate for clinical outcome variables.

D. Beyond Clinical Medicine: Transferable Decision-Support Architectures

The causal BN decision-support paradigm developed for medicine generalizes naturally to other high-stakes, evidence-based decision domains, and [85] provide a concrete instance in judicial decision-making. Their system structures legal evidence using conditional probability tables and historical case data, dynamically updating outcome probabilities as new evidence is introduced—mirroring exactly the sequential belief-updating process described for diagnosis in [83] and the evidence-fusion role described for radiology in [82]. The reported emphasis on transparency, reduced cognitive overload, and bias mitigation echoes the interpretability arguments made throughout the healthcare BN literature, reinforcing that the core value proposition of causal BNs—auditable, evidence-updating, human-interpretable probabilistic reasoning—is domain-agnostic even though the present survey's primary interest is medical application.

Across these strands, a clear arc emerges: early and survey-level work ([8], [50], [80], [82], [1]) established the conceptual and practical case for BNs in medicine relative to opaque machine-learning alternatives, while documenting shallow and uneven real-world adoption; a middle layer of work ([81], [83], [86]) adapted BN inference to the temporal and evidentially-updating structure of diagnosis; and a recent, largely theoretical wave ([4], [41], [36], [42], [87], [84]) has pushed toward automated causal structure learning, counterfactual reasoning, streaming adaptation, and decision-theoretic integration, with [84] and [85] showing that these ideas transfer productively to adjacent applied decision-making domains.

Overall, causal BNs occupy a distinctive niche in medical AI: they are consistently championed for interpretability, bidirectional reasoning, and principled fusion of data with prior clinical knowledge, and their value is reinforced by applied demonstrations in policy and legal decision-support that share medicine's need for auditable, evidence-updating inference. Yet the field remains fragmented—adoption is concentrated in a few medical conditions [50], much of the causal-discovery methodology developed in 2026-era work is presented without empirical clinical validation, and direct head-to-head comparisons against deep learning or hybrid neuro-symbolic pipelines [82] remain scarce. Closing this gap between methodological innovation in causal structure learning and rigorous clinical validation—ideally on shared, realistic healthcare datasets with explicit handling of confounding, temporal dynamics, and streaming updates—stands out as the principal open challenge for this line of research.

VII. Structure Learning and Conditional Independence Foundations

Bayesian networks derive their interpretive and computational power from the factorization of a joint distribution according to a directed acyclic graph (DAG) that encodes conditional independence relationships among variables. Before any inference or decision-support application can be built, two foundational questions must be addressed: what structural assumptions make inference tractable, and how can the DAG itself be recovered from data or domain knowledge. This section traces the evolution of this line of work, from early theoretical results on exploiting structural regularities for efficient inference, through the maturation of constraint-based and score-based structure learning algorithms, to the software ecosystems and applied studies that have tested these methods' robustness and usability in realistic settings.

A. Causal Independence and Inference Complexity as Structural Priors

A central insight that shaped subsequent structure learning research is that the conditional probability tables attached to a DAG are rarely arbitrary; they often exhibit additional regularities, such as causal independence, that can be exploited both for more efficient inference and for more parsimonious structure specification. [24] formalized this by showing that when the conditional probability of a variable given its parents can be expressed via an associative and commutative combination operator (e.g., or, sum, max) over the independent contributions of each parent, the joint distribution admits a finer-grained factorization than the naive conditional probability table. Building on the variable elimination (VE) algorithm, they demonstrated on the CPCS medical diagnosis networks that exploiting causal independence yields substantially more efficient exact inference and allows networks larger than previously tractable to be handled, directly linking

TABLE IX
SUMMARY OF WORKS ON STRUCTURE LEARNING AND CONDITIONAL INDEPENDENCE FOUNDATIONS

Ref.	Year	Venue	Main contribution
[24]	1996	Journal of Artificial Intelligence Res...	Exploits causal independence to factorize conditional probabilities for faster Bayesian inference
[88]	1997	Artificial Intelligence	Proves efficient randomized approximation algorithm for Bayesian inference without extreme probabilities
[9]	2012	Journal of Climate	Applies constraint-based causal structure learning to climate low-frequency variability modes
[46]	2013	arXiv	Explains why belief network diagnosis performance is insensitive to probability imprecision
[3]	2016	Bioinformatics	R package learning Bayesian network structure and parameters with missing data
[89]	2016	Knowledge-Based Systems	Parallelizes conditional independence testing in the PC algorithm for large-scale structure learning
[47]	2021	Journal of Computational and Graphical...	Hybrid MCMC-constraint method for efficient Bayesian network structure sampling and learning
[90]	2021	International Journal of Approximate R...	Empirically validates 15 Bayesian network structure learning algorithms under noisy real-world-like data
[13]	2022	Artificial intelligence in medicine	User study evaluating understandability of explanation methods for Bayesian network predictions
[91]	2022	Mathematics	Describes OpenMarkov software as pedagogical tool for teaching probabilistic graphical models
[15]	2023	Artificial Intelligence Review	Comprehensively surveys 74 combinatoric algorithms for learning Bayesian network structure from data
[92]	2023	International Journal of Intelligent I...	Uses dynamic Bayesian networks with rough set reduction for airborne electronics fault diagnosis
[93]	2024	Journal of Imaging	Compares explainable Bayesian network versus black-box model for hospital admission prediction

assumptions about local structure (causal independence) to global inference complexity. Complementary to this exact-inference result, [88] addressed the worst-case hardness of approximate inference, proving that even approximating posterior probabilities is NP-hard in general but constructing a randomized bounded-variance algorithm—extending likelihood weighting—that achieves provably efficient approximation whenever conditional probabilities are not extreme. They further derandomized this scheme using pseudorandom generators, obtaining subexponential worst-case time bounds as a function of network depth, in contrast to the exponential behavior of earlier deterministic approximations. Together, these two papers established that both exact and approximate inference tractability are governed by structural properties of the network (causal independence, extremity of probabilities, depth), setting expectations for what structure learning algorithms should aim to recover or preserve.

A further foundational observation concerns the sensitivity of inferred structure and parameters to imprecision. [46] examined belief networks for diagnosis (subsets of the CPCS and QMR networks) and found that diagnostic performance, measured as average posterior probability assigned to the true disease, was surprisingly insensitive to substantial random noise injected into conditional probabilities. They attributed this robustness partly to the use of posterior probability as a symmetric-noise-insensitive performance criterion and partly to the fact that true posteriors tend to cluster near zero or one. This result has direct implications for structure learning: if diagnostic utility is largely insensitive to parameter imprecision, then errors introduced during structure or parameter estimation—an unavoidable consequence of finite, noisy data—may matter less for downstream decision quality than classical structural-accuracy metrics would suggest, a theme later revisited by empirical structure-learning validation studies.

B. Constraint-Based, Score-Based, and Hybrid Structure Learning Algorithms

The dominant algorithmic paradigms for recovering BN structure from data are constraint-based methods, which test for conditional independence to prune edges, and score-and-search methods, which explore the space of DAGs using a scoring criterion. [9] provide an accessible exposition of constraint-based structure learning for a non-specialist (climate science) audience and apply it to recover temporal and static independence graphs among modes of atmospheric low-frequency variability (WPO, EPO, PNA, NAO), illustrating both the interpretive value of learned Bayesian networks as causal hypotheses and the domain-specific validation challenges of such methods when ground truth is unavailable and only indirect physical plausibility can be invoked. Scaling constraint-based learning to large datasets is itself a nontrivial engineering problem: [89] identify the conditional independence testing phase of the PC algorithm as the dominant computational bottleneck and propose a parallel variant that distributes this step, demonstrating performance gains across five real-world network topologies compared to both the serial PC algorithm and an alternative process-based parallelization scheme, though the approach remains fundamentally constrained by the correctness assumptions (faithfulness, causal sufficiency) underlying the PC algorithm itself.

Hybrid approaches attempt to combine the computational economy of constraint-based tests with the statistical guarantees of score-based search. [47] propose an MCMC scheme in which each sampling step reduces to simple table lookups derived from conditional independence constraints, together with an iterative correction procedure to mitigate errors from imperfect independence tests; this yields long, efficiently sampled Markov chains over DAG space and enables full Bayesian model averaging at scales previously prohibitive for MCMC-based structure learning, reportedly outperforming existing alternatives. The accumulation of such diverse algorithmic

families, however, has made comparative assessment difficult, since different papers use inconsistent evaluation protocols and often rely on synthetic data satisfying their own assumptions. [15] respond to this fragmentation with a comprehensive review of 74 structure learning algorithms, unifying their descriptions in common terms, cataloguing similarities and differences, and critically examining the consistency (or lack thereof) of comparative performance claims across the literature, alongside methods for handling noisy data and incorporating expert knowledge. In a complementary empirical study, [90] directly interrogate the gap between synthetic and real-world performance by applying 15 established and recent structure learning algorithms to data perturbed with synthetic noise across multiple case studies, sample sizes, and evaluation criteria—an effort spanning roughly 10,000 learned graphs and seven months of computation—concluding that algorithm rankings and absolute performance are highly sensitive to data quality assumptions such as completeness and causal sufficiency, reinforcing the concern (anticipated conceptually by the robustness findings of [46]) that idealized synthetic benchmarks can substantially overstate real-world effectiveness.

C. Software Ecosystems for Structure Learning and Dissemination

Beyond algorithmic development, the practical uptake of BN structure learning has depended on accessible software implementations that handle realistic data imperfections and support pedagogical use. [3] introduce an open-source R package that, unusually among contemporaneous tools, integrates structure and parameter learning with inference under missing data, addressing a gap in the available open tooling for a problem that is pervasive in practical datasets but often sidestepped by algorithms assuming complete data. On the educational side, [91] describe OpenMarkov, a widely adopted open-source tool used in over 30 countries, and demonstrate how its interactive structure-learning facilities—implementing both search-and-score and PC-style algorithms—can be used to teach core concepts such as conditional independence, d-separation, and Markov blankets, highlighting the role of tooling not only in research but in disseminating the conceptual foundations of Bayesian network structure learning to new practitioners.

D. Applied Structure Learning for Diagnosis and Decision Support

A further strand of work evaluates learned or expert-constructed Bayesian network structures in applied diagnostic settings, probing whether structural and explanatory properties translate into practical utility. [92] apply dynamic Bayesian networks to fault diagnosis of airborne electronic equipment, incorporating rough set theory to reduce non-essential fault-symptom attributes before structure specification, illustrating how dimensionality reduction can be coupled with dynamic BN structures in safety-critical industrial diagnosis. In the clinical domain, [93] compare an explainable-by-design Bayesian network against a black-box Random Forest for supporting triage decisions during the COVID-19 pandemic, finding comparable predictive performance while emphasizing the BN's advantage in generalizing across pandemic waves and in offering transparent reasoning to clinicians—an empirical argument for structure learning approaches that preserve interpretability. Complementing this performance-oriented evaluation, [13] shift focus to the human factors of BN-based explanation, surveying clinicians' perceived understanding across four different explanation approaches for Bayesian network inference and finding that technical soundness of an explanation method does not guarantee favorable user experience, underscoring that the value of a learned BN structure in practice depends jointly on its statistical fidelity and its comprehensibility to end users.

Collectively, this body of work shows a clear trajectory: early theoretical results established that structural regularities (causal independence, bounded extremity of probabilities) govern the tractability of both exact and approximate inference, which in turn motivated algorithms explicitly designed to recover or exploit such regularities from data. The field subsequently diversified into constraint-based, score-based, and hybrid structure learning algorithms, with increasing attention to scalability (parallelization), robustness to missing or noisy data, and rigorous comparative evaluation, as crystallized in the large-scale survey and empirical validation efforts. Software tools have played a supporting but essential role in making these advances usable and teachable, while applied studies in clinical and industrial diagnosis demonstrate that structural learnability and inferential robustness do not automatically guarantee user trust or comprehensibility. Remaining gaps include the continued reliance on synthetic benchmarks that may not reflect real-world violations of causal sufficiency and completeness, the relative scarcity of standardized, reproducible evaluation protocols across algorithm families, and a persistent disconnect between algorithmic performance metrics and the human-centered explainability requirements increasingly demanded of BN-based AI systems in high-stakes domains.

VIII. Exact and Approximate Inference Algorithms

Bayesian networks owe much of their practical appeal to the existence of efficient algorithms for computing posterior marginals and most probable explanations, yet exact inference is #P-hard in general and tractable only when the underlying graph has low treewidth. This tension between exactness and scalability has driven a long line of research on approximate inference, which trades guaranteed correctness for computational feasibility on dense, hybrid, or very large networks. As the historical synthesis in [49] traces, the field evolved from Pearl's tree-structured message-passing engines and the junction-tree algorithm of local computations toward belief propagation on general graphs and variational methods that scale inference beyond tree structure; the works surveyed below occupy different points along this trajectory, from refinements of classical message passing and conditioning to hybrid-network methods, neural amortized inference, and domain-specific deployments.

TABLE X
SUMMARY OF WORKS ON EXACT AND APPROXIMATE INFERENCE ALGORITHMS

Ref.	Year	Venue	Main contribution
[30]	2012	arXiv	Derives convergent double-loop expectation propagation algorithm for approximate inference in dynamic Bayesian networks
[94]	2012	arXiv	Presents two approximate inference algorithms for hybrid Bayesian networks with discrete constraints
[26]	2013	arXiv	Empirically compares loopy belief propagation marginals to exact inference in Bayesian networks
[25]	2013	arXiv	Dynamic conditioning and B-conditioning algorithms for exact and approximate causal network inference
[27]	2013	arXiv	Expectation Propagation unifies assumed-density filtering and loopy belief propagation for hybrid networks
[95]	2013	arXiv	Presents search-based approximate inference algorithm for arbitrary noisy-OR belief networks
[28]	2014	arXiv	Feedforward inference network enables fast variational training of sigmoid belief networks
[96]	2017	ACM Transactions on Privacy and Securi...	Applies Loopy Belief Propagation for scalable approximate Bayesian inference on attack graphs
[97]	2017	IEEE Transactions on Industrial Inform...	Proposes fuzzy probability Bayesian network for dynamic industrial control system cybersecurity risk assessment
[29]	2018	Journal of Artificial Intelligence Res...	Reviews inference algorithms for hybrid discrete-continuous Bayesian networks
[49]	2026	arXiv	Narrative review tracing probabilistic graphical models from inference engines to causality

A. Message Passing and Conditioning Methods

The most influential line of approximate inference research treats loopy belief propagation (LBP) as a heuristic extension of Pearl's exact polytree algorithm to graphs with cycles. The empirical study in [26] was foundational in establishing that LBP, despite having no general correctness guarantee, often produces marginals close to the exact posteriors on real-world networks such as ALARM, while on other networks such as QMR the beliefs can oscillate and bear little relation to the true posteriors; the authors further show that naive fixes for oscillation can make results worse, highlighting that empirical success of LBP is case-dependent and not yet well understood theoretically. This motivated more principled reformulations: [27] unifies assumed-density filtering and loopy belief propagation under the umbrella of Expectation Propagation (EP), which approximates beliefs by low-order moments (e.g., mean and variance) rather than full distributions, enabling application to hybrid discrete-continuous networks and richer, correlation-preserving messages; EP is shown to outperform Laplace's method, variational Bayes, and Monte Carlo on Gaussian mixture inference and to support Bayes point machine training. [30] extends EP specifically to dynamic Bayesian networks and supplies the convergence theory that the original formulation lacked: although greedy EP need not converge, the authors derive a double-loop algorithm guaranteed to converge to a local minimum of the Bethe free energy and show that stable fixed points of (damped) EP correspond to such minima, though the converse does not hold — a result that also clarifies when LBP-style methods on general Bayesian networks can be trusted.

Complementary to message-passing, conditioning-based methods attack intractability by decomposing inference into simpler sub-problems via variable instantiation. [25] revisits cutset conditioning, proposing dynamic conditioning, which achieves linear complexity on networks where classical cutset conditioning remains exponential, and B-conditioning, an approximate variant that explicitly trades approximation quality for computation time. Unlike LBP and EP, which operate via iterative local updates, these conditioning algorithms are recursive and offer more predictable complexity control, illustrating that the exact/approximate divide in Bayesian network inference is not a single axis but a design space combining message passing, moment matching, and structured conditioning, each with different convergence and accuracy trade-offs.

B. Inference in Hybrid and Structured Networks

A second strand of work targets networks that combine discrete and continuous variables, or that exploit special local structure, where standard junction-tree or sum-product algorithms are especially costly. [94] addresses Hybrid Mixed Networks, which augment hybrid Bayesian networks with explicit discrete deterministic constraints, by combining Generalized Belief Propagation, Rao-Blackwellised importance sampling, and constraint propagation; the resulting algorithms are evaluated on randomly generated HMNs, showing that integrating constraint reasoning with sampling and message passing can address complexity that neither technique handles well alone, though the approach is validated only on synthetic instances rather than real-world hybrid models. [95] instead exploits a different structural regularity — the noisy-OR interaction model — proposing TopEpsilon, a search-based approximate inference algorithm that generalizes earlier two-level noisy-OR techniques to arbitrary noisy-OR networks; while promising on initial experiments, its applicability is inherently limited to networks whose conditional probability tables admit the noisy-OR factorization. Bridging these specialized techniques, the review of [29] systematically organizes inference approaches for hybrid Bayesian networks (including Gaussian, mixture, and other parametric families) by their methodological basis, extends the discussion to hybrid dynamic Bayesian networks, and surveys the software ecosystem supporting such inference; it makes clear that, unlike the discrete case, hybrid-network inference rarely admits closed-form solutions, so the methods compared in [94] and [95] represent instances of a broader taxonomy rather than isolated solutions.

C. Neural and Learning-Based Approximate Inference

A more recent departure replaces iterative numerical approximation with learned, amortized inference. [28] trains a feedforward inference network jointly with a directed generative model (sigmoid belief networks and deep autoregressive networks) by maximizing a variational lower bound, using model-independent variance-reduction techniques to make the otherwise high-variance gradient estimator practical. This yields fast, non-iterative, exact sampling from an approximate posterior at test time, in contrast to the iterative message-passing or conditioning schemes above, and the method is shown to outperform the classical wake-sleep algorithm on MNIST and to achieve state-of-the-art results on the Reuters RCV1 dataset. This line of work signals a shift away from hand-designed approximate inference toward inference procedures that are themselves learned, foreshadowing the integration of Bayesian network inference with deep generative modeling.

D. Scaling Approximate Inference for Risk and Security Applications

A final cluster of work demonstrates how generic approximate inference machinery is adapted to large, application-specific Bayesian networks where exact inference is infeasible. [96] applies loopy belief propagation to Bayesian attack graphs for security risk assessment, showing both sequential and parallel variants scale linearly in the number of nodes for static and dynamic analysis — in contrast to exact techniques, which do not scale — and validating accuracy and convergence on synthetic attack graphs with thousands of nodes and varying topologies/clustering. [97] targets a different obstacle to inference in industrial control system cybersecurity: insufficient historical data to populate crisp conditional probabilities. The authors replace point probabilities with fuzzy probabilities in a fuzzy probability Bayesian network and design an approximate dynamic inference algorithm augmented with a noise-evidence filter to suppress spurious evidence from system faults, demonstrating the approach on a simplified chemical reactor testbed. Together these studies show that approximate inference in practice is rarely applied off-the-shelf; it is typically combined with domain-specific representational choices (constraint handling, fuzzy probabilities, graph clustering) to meet both scalability and robustness requirements, though evaluation in both cases remains confined to synthetic or simplified systems rather than deployed real-world infrastructures.

Taken together, this body of work charts a clear progression: from theoretically grounded refinements of exact conditioning and message passing [25], [26], through moment-based generalizations such as EP that extend inference to dynamic and hybrid settings with partial convergence guarantees [27], [30], to specialized hybrid- and structure-exploiting algorithms [29], [94], [95], learned amortized inference [28], and finally large-scale domain deployments [96], [97], all situated within the broader historical arc described in [49]. Despite this progress, important gaps remain: convergence and accuracy guarantees for loopy and expectation-propagation methods are still incomplete outside restricted settings, evaluation is dominated by synthetic benchmarks or small canonical networks rather than large real-world graphs, and there is no unified framework bridging classical message-passing/conditioning methods with learned neural inference for general hybrid, dynamic, and structured Bayesian networks — leaving the accuracy-scalability trade-off an open problem as networks grow in size and heterogeneity.

IX. Metaheuristic Optimization for Structure Learning

Learning the structure of a Bayesian network from data is formally an NP-hard combinatorial optimization problem: the space of directed acyclic graphs (DAGs) over n variables grows super-exponentially, and exhaustive search-and-score over this space is intractable beyond small n. This has made structure learning a natural testbed for metaheuristic optimization, where exact or greedy algorithms are replaced by stochastic, often nature-inspired, search procedures that trade optimality guarantees for scalability and empirical performance. The literature can be organized along two main evolutionary threads: hybrid approaches that combine classical local/constraint-based pruning with greedy or heuristic refinement, and a more recent wave of swarm- and bio-inspired metaheuristics (pigeons, bees, falcons, bacteria) together with evolutionary algorithms that reformulate structure learning as a general combinatorial optimization problem to be solved with population-based search.

A. Hybrid Local-Global and Instance-Specific Search Strategies

The earliest and most influential attempt to tame the search space while retaining strong empirical accuracy is Max-Min Hill-Climbing (MMHC) [14], which decouples structure learning into a constraint-based skeleton identification phase (restricting the candidate parent sets of each node) followed by a Bayesian-scoring greedy hill-climbing search that only orients edges within this restricted skeleton. By combining local learning, constraint-based, and search-and-score paradigms, MMHC was shown in extensive comparisons to outperform PC, Sparse Candidate, Three Phase Dependency Analysis, Optimal Reinsertion, and Greedy Equivalence Search (GES), establishing it as a long-standing baseline against which later heuristic and metaheuristic methods are routinely benchmarked. Building on this search-and-score philosophy but shifting the goal from population-level to individual-level models, [98] proposes an instance-specific structure learning method that guides the search using the attributes of a particular case (e.g., a patient's symptoms and lab results), reporting improved precision over FGES—an implementation of GES—especially for variables exhibiting context-specific independence; this illustrates that the same greedy search-and-score machinery can be repurposed for personalized rather than global network discovery. More recently, [100] targets the efficiency bottleneck of hill-climbing itself, proposing a random-forest-inspired probabilistic incremental measure to both construct an informed initial structure and constrain the

TABLE XI
SUMMARY OF WORKS ON METAHEURISTIC OPTIMIZATION FOR STRUCTURE LEARNING

Ref.	Year	Venue	Main contribution
[14]	2006	Machine Learning	Combines local learning and greedy search for Max-Min Hill-Climbing structure learning
[98]	2018	arXiv	Introduces instance-specific Bayesian network structure learning guided by individual patient attributes
[18]	2019	Journal of Cognitive Science	Applies Pigeon Inspired Optimization algorithm to Bayesian network structure learning
[20]	2019	Information	Hybrid optimization algorithm improving Bayesian network structure learning accuracy and efficiency
[99]	2021	Computer Science	Applies Falcon Optimization Algorithm to learn Bayesian network structures
[19]	2022	International Journal of Intelligent S...	Improves genetic algorithm with local optimization decoder for Bayesian network structure learning
[100]	2022	IEEE Access	Constrains hill-climbing search using probabilistic incremental node association analysis for structure learning
[23]	2023	International Journal of Interactive M...	Falcon Optimization Algorithm proposed for learning Bayesian network structures
[21]	2024	Scientific Reports	Improves bacterial foraging optimization algorithm for learning dynamic Bayesian network structures

search space via node-centered probability-increment thresholds, after which a "forbidden" hill-climbing search with BIC scoring refines the result; the reported gains are chiefly in runtime, with comparable or better structural quality than conventional hill-climbing variants. Collectively, this line of work shows that intelligent pre-structuring of the search space—whether via constraint-based skeletons, instance-level guidance, or association-based priors—remains a powerful complement to, rather than a replacement for, local greedy search.

B. Swarm and Nature-Inspired Metaheuristics

A second, more recent thread abandons greedy local search almost entirely in favor of population-based swarm intelligence operating directly over candidate DAGs scored with BDeu or similar metrics. [18] introduces Pigeon-Inspired Optimization (PIO), which evolves a population of DAGs using map-and-compass and landmark operators analogous to pigeon navigation, and reports superior BDeu scores and classification performance relative to simulated annealing, bee-optimization hybrids, and greedy search on benchmark networks. This established a template—population of graphs, nature-inspired move operators, confusion-matrix and score-based evaluation against greedy/annealing baselines—that subsequent papers largely replicate with different biological metaphors. [20] discretizes an artificial bee colony algorithm and hybridizes it with complementary optimization steps to address the low precision/high complexity trade-off of pure data-driven structure learning, showing improved structure quality and convergence on simulated data. [99] proposes the Falcon Optimization Algorithm (FOA), using falcon prey-search dynamics together with standard graph-edit operations (reversing, deleting, moving, inserting edges) to navigate the DAG space, and reports better BDeu scores and accuracy than PIO, greedy search, and simulated annealing; [23] extends this evaluation of FOA against PIO, greedy search, and antlion optimization across further benchmark datasets, again finding FOA favorable in both score and confusion-matrix metrics. The consistent pattern across this PIO/FOA family is that each new metaheuristic is validated primarily against the immediately preceding one plus classic greedy/simulated-annealing baselines, rather than against stronger hybrid methods such as MMHC, leaving open how competitive these swarm algorithms are relative to the broader structure-learning literature. Extending swarm intelligence to the temporal setting, [21] proposes an improved bacterial foraging optimization algorithm (IBFO-A) for learning Dynamic Bayesian Network structures, incorporating chaotic-map initialization, an osprey-inspired exploration strategy, genetic-style multi-point crossover for colony propagation, and an elimination-dispersal step, reporting improved generation efficiency and structural accuracy on DBN benchmarks; this reflects a trend toward hybridizing multiple metaheuristic mechanisms within a single algorithm to balance exploration and exploitation, at the cost of increasingly complex, multi-stage frameworks whose individual component contributions are harder to isolate.

C. Evolutionary and Genetic Algorithms

Complementing swarm intelligence, evolutionary algorithms treat structure learning as a combinatorial optimization problem solved via selection, crossover, and mutation over encoded graph representations. [19] improves a biased random-key genetic algorithm for Bayesian network structure learning, pairing it with a local optimization model as a decoder to translate genetic encodings into valid, high-quality DAGs; evaluated on nine benchmark networks and a real cross-site-scripting attack dataset, the method is reported to yield more accurate structures than other state-of-the-art algorithms and to perform well in a downstream web-security detection task, suggesting that genetic encodings combined with a problem-specific local decoder can mitigate the risk of generating invalid (cyclic) structures that plagues naive evolutionary search over DAG space.

Across these three groupings, a clear trajectory emerges: from principled hybridizations of constraint-based and greedy score-based search (MMHC and its instance-specific and efficiency-oriented descendants), toward an expanding menagerie of nature-inspired swarm metaheuristics (pigeons, bees, falcons, bacteria) that largely validate themselves against greedy search, simulated annealing, and each other, and finally to genetic-algorithm formulations that explicitly manage DAG validity through specialized decoders. A recurring limitation is

the lack of standardized, head-to-head benchmarking against strong hybrid baselines like MMHC or GES-based methods under identical scoring functions and datasets; most swarm-based papers compare mainly within their own lineage (PIO vs. FOA vs. antlion), making cross-family comparisons and claims of overall state-of-the-art status difficult to assess. Scalability to high-dimensional variable sets, sensitivity to hyperparameters of the bio-inspired operators, and extension beyond static networks (as only partially addressed by dynamic BN work) remain open challenges for this line of research.

X. Computational Complexity of Structure Learning and Inference

Learning the structure of a Bayesian network from data and subsequently performing inference over it are both computationally hard problems, and this intrinsic hardness has shaped almost every algorithmic development in the field. Exact inference in general Bayesian networks is #P-hard, and [31] showed that even approximate inference—long assumed to be tractable—is NP-hard in the general case, closing off the hope that relaxing exactness alone would yield efficient algorithms. Structure learning compounds this difficulty: the number of possible directed acyclic graphs (DAGs) over n variables grows super-exponentially, and [48] frames the resulting combinatorial search as the central open challenge common to all probabilistic graphical model learning, surveying the landscape of score-based, constraint-based and hybrid strategies that have emerged to cope with it. Against this backdrop, the subsequent literature bifurcates along two lines: work that seeks exact or provably optimal solutions at high computational cost, and work that trades optimality for scalability through heuristics, approximations, or alternative inference paradigms altogether.

A. Hardness of Exact and Approximate Inference

The foundational result of [31] established that relative approximation of conditional probabilities in belief networks is as hard as exact computation unless NP $\subseteq$ P, meaning that polynomial-time approximation schemes cannot exist for the general class of networks without further structural restrictions (e.g., bounded treewidth or bounded node in/out-degree). This result reframed decades of inference algorithm design: methods such as likelihood weighting or Gibbs sampling can only be guaranteed efficient under restricted topologies or error tolerances, not in general. A markedly different response to the intractability of exact Bayesian inference has emerged from the deep learning literature, where [44] recast dropout training of convolutional neural networks as a form of approximate variational inference in Bayesian neural networks. By using Bernoulli variational distributions over convolutional kernels, this approach avoids explicit posterior computation altogether, achieving practical scalability (demonstrated on CIFAR-10) at the cost of being only a loose, implicit approximation to a true Bayesian posterior. Juxtaposing [31] and [44] illustrates two opposite strategies for confronting inference hardness: the former characterises worst-case limits for exact probabilistic graphical models, while the latter sidesteps the issue by embedding approximate Bayesian reasoning inside a differentiable, likelihood-free training pipeline—sacrificing theoretical guarantees for empirical tractability at scale.

B. Exact Structure Learning: Dynamic Programming, Integer Programming, and Constraint Programming

A second cluster of work tackles the NP-hardness of Bayesian network structure learning (BNSL) head-on, seeking exact, optimal solutions for decomposable scores such as BDeu or BIC rather than resorting to heuristics. [16] propose a dynamic-programming algorithm that reduces the super-exponential cost of exhaustive DAG enumeration to a "merely" exponential space/time procedure, guaranteeing a global optimum but remaining limited to small-to-moderate numbers of variables due to the exponential memory footprint inherent to dynamic programming over subsets of nodes. [17] instead formulate BNSL as an integer programming (IP) problem, as implemented in the GOBNILP system, and contribute a deeper theoretical analysis of the polytopes underlying this formulation: they prove that the associated cutting-plane separation problem is itself NP-hard, and characterise the facets of three key polytopes, providing both theoretical grounding and practical guidance for improving IP-based solvers. Building on this cutting-plane tradition, [33] introduce a constraint-programming (CP) solver, CPBayes, that uses a polynomial-time algorithm for discovering a useful subset of cluster cuts together with a generalised arc-consistency propagator for acyclicity; although the LP relaxation is solved only approximately via a greedy algorithm, the resulting solver improves performance by orders of magnitude over naive CP formulations and compares favourably with GOBNILP, showing that acyclicity-aware constraint propagation can substitute, at lower per-cut cost, for the exact but expensive separation routines of pure IP approaches. Collectively, these three works trace an evolution from brute-force dynamic programming, to IP with strong polyhedral theory, to CP methods that exploit cheaper, approximate reasoning about acyclicity and cluster cuts to scale exact or near-exact BNSL further, though all remain fundamentally constrained by the superexponential size of the DAG search space and are typically only practical for datasets with tens, rather than hundreds, of variables.

C. Heuristic Search, Metaheuristics, and Practical Sensitivities

Given the limits of exact approaches, a parallel body of work accepts suboptimality in exchange for scalability, while also scrutinising the practical robustness of such heuristic algorithms. [22] propose OP-PSO-DE, a metaheuristic combining particle swarm optimisation, differential evolution, and opposition-based learning to narrow the search space explored by population-based search, reporting improved performance over other state-of-the-art structure learning algorithms at a fixed sample size of 500; as with most metaheuristic approaches, however, convergence to the global optimum is not guaranteed and performance is sensitive to algorithm-specific hyperparameters. Complementing this algorithmic contribution, [32] provide a critical empirical study of a confound

TABLE XII
SUMMARY OF WORKS ON COMPUTATIONAL COMPLEXITY OF STRUCTURE LEARNING AND INFERENCE

Ref.	Year	Venue	Main contribution
[31]	1993	Artificial Intelligence	Proves approximate probabilistic inference in Bayesian belief networks is NP-hard
[48]	2011	arXiv	Comprehensive survey of structure learning methods for probabilistic graphical models
[44]	2015	arXiv	Approximates Bayesian inference in CNNs using Bernoulli variational dropout distributions
[17]	2017	Journal of Artificial Intelligence Res...	Analyzes integer programming polytopes and complexity for optimal Bayesian network structure learning
[16]	2018	arXiv	Presents an exponential-time dynamic programming algorithm for finding globally optimal Bayesian network structures
[33]	2021	arXiv	Improves acyclicity reasoning in constraint programming solver for Bayesian network structure learning
[32]	2024	Data Mining and Knowledge Discovery	Shows variable ordering strongly affects Bayesian network structure learning algorithms
[22]	2025	Scientific Reports	Combines particle swarm and differential evolution with opposition-based learning for BN structure learning

that is rarely addressed in the structure learning literature: the sensitivity of learnt graphs to the arbitrary ordering in which variables are presented to the learning algorithm. They show that hill-climbing and its variants are particularly affected, but that the effect extends, to a lesser degree, to hybrid and constraint-based algorithms as well, raising concerns about the validity of published performance comparisons—including, implicitly, many heuristic and metaheuristic methods such as [22]—that do not control for variable ordering. This juxtaposition highlights a gap between algorithmic innovation aimed at improving search efficiency and the methodological rigor needed to evaluate such innovations reliably.

Taken together, these three threads reveal a field still negotiating a fundamental trade-off between optimality and scalability. The theoretical hardness results of [31] and the polyhedral complexity analysis of [17] establish firm lower bounds on what can be achieved efficiently, both for inference and for the structure-learning subproblems (e.g., separation) that underlie exact solvers. Exact methods [16], [17], [33] have made steady algorithmic progress—from dynamic programming, to IP with strong cutting planes, to CP with acyclicity-specific propagation—but remain bounded by the same super-exponential search space identified early on by [48], limiting their applicability to networks of modest size. Heuristic and metaheuristic methods [22] extend applicability to larger problems but at the cost of optimality guarantees, and the methodological finding of [32] that variable ordering alone can dominate performance differences suggests that many reported comparisons across heuristic algorithms may be confounded. Meanwhile, approximate Bayesian deep learning approaches such as [44] largely abandon the structure-learning and exact-inference machinery of classical Bayesian networks altogether, instead embedding Bayesian-flavoured uncertainty estimation into scalable gradient-based training—raising the question of how insights from classical BNSL hardness results might inform, or be reconciled with, these more statistically loose but highly scalable neural alternatives.

In summary, the computational complexity of Bayesian network structure learning and inference remains a binding constraint: foundational hardness results rule out efficient exact or approximate general-purpose algorithms, exact solvers have advanced considerably but still scale poorly with dimensionality, and heuristic alternatives that promise scalability are only beginning to be scrutinised for confounds such as variable ordering. Open gaps include principled hybridisation of exact polyhedral/constraint techniques with heuristic search to extend exact guarantees to larger domains, standardised evaluation protocols that control for ordering and other arbitrary input artefacts, and a clearer theoretical bridge between classical complexity results for discrete BNs and the approximate, variational notions of Bayesian inference now used in large-scale neural architectures.

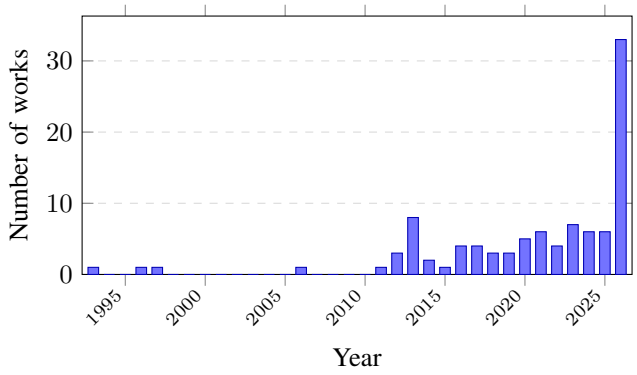


Fig. 1. Distribution of the surveyed works by publication year.

XI. DISCUSSION

A. Cross-Cutting Trends in the Literature

The temporal distribution of the surveyed literature, summarized in Fig. 1, spans more than three decades (1993–2026) and reveals a markedly non-uniform growth pattern: foundational theoretical contributions from the 1990s and 2000s [14], [24], [31] are followed by a long period of steady methodological consolidation, before an sharp acceleration in the last three years, which account for 45% of the 100 works reviewed and a peak in 2026. This recent surge is dominated by applications of dynamic and causal Bayesian networks to streaming, temporal, and high-dimensional data [34], [35], [38], [58], [60], [69], [72], as well as by hybridizations with neural, symbolic, and quantum-inspired paradigms [59], [63]–[65]. This pattern suggests that Bayesian networks, far from being a legacy

formalism, are being actively repurposed as a substrate for integrating probabilistic reasoning with modern AI architectures, particularly in response to demands for interpretability and causal grounding that purely statistical deep learning models struggle to satisfy [1], [4], [49].

B. Convergence of Causal Discovery and Deep Representation Learning

A dominant cross-cutting theme is the convergence between classical Bayesian network structure learning and deep representation learning. Early approximate-inference and structure-learning work established the algorithmic backbone—constraint-based search [14], [89], score-based dynamic programming [16], integer programming [17], and MCMC-based sampling [47]—which later surveys and validations systematized and benchmarked [15], [32], [48], [90]. The recent wave of causal-discovery papers builds directly on this foundation but increasingly substitutes or augments these combinatorial searches with neural components: variational inference for joint structure-and-parameter learning [35], [67], attention-based convolutional networks for temporal causal graphs [51], deep Bayesian network learners [65], and episodic-memory or graph-neural architectures [66], [68], [70]. This mirrors the broader trend identified in works that explicitly bridge graphical models and neural networks for structured inference [28], [43], [44], [76], and in generative approaches to Bayesian computation [45]. Temporal extensions—dynamic Bayesian networks, Hidden Markov Model hybrids, and Granger-causality integrations [36], [37], [39], [40], [53], [54], [62], [81]—constitute perhaps the single most populous sub-theme, reflecting sustained interest in modeling evolving causal structure rather than static graphs, a concern also present in metaheuristic structure-learning work targeting dynamic networks [21].

C. Dominant Techniques, Application Domains, and Benchmarking Practices

Across themes, three technique families recur: (i) search-and-score or constraint-based structure learning, increasingly hybridized with metaheuristics such as genetic algorithms, pigeon- and falcon-inspired optimization, bacterial foraging, and opposition-based learning [18]–[20], [22], [23], [99], [100]; (ii) exact and approximate inference algorithms—belief propagation, expectation propagation, conditioning methods, and hybrid discrete-continuous solvers—developed primarily in the 1990s–2010s and still cited as the computational backbone for newer applications [25]–[27], [29], [30], [88], [94], [95]; and (iii) domain-specific causal or diagnostic models, most visibly in medicine, where Bayesian and causal Bayesian networks are used for diagnosis, prognosis, and explanation [5]–[7], [12], [50], [73], [82], [83], [86], [93], as well as in neuroscience [8], climate science [9], cybersecurity [96], [97], fault diagnosis [92], judicial decision support [85], and policy-making [84]. Despite this diversity of application domains, evaluation practices remain strikingly heterogeneous and often weakly standardized: many structure-learning papers rely on classical benchmark networks (e.g., well-known discrete graphs used across [15], [32], [33], [90]), while the 2026 wave of causal-discovery papers more frequently reports on synthetic or simulated streaming/time-series data with less consistent baselines, making cross-paper comparison difficult. Where human-centered evaluation is attempted, it is rare and qualitative, as in the user study of explanation understandability for Bayesian network predictions [13] or the comparison of explainable versus black-box models for pandemic-era hospital admission prediction [93].

D. Reproducibility, Tooling, and Theoretical Grounding

Reproducibility is unevenly supported across the literature. A relatively small number of works contribute openly usable software and pedagogical infrastructure—bnstruct for structure learning with missing data [3], RevBayes for phylogenetic graphical models [52], and OpenMarkov as a teaching platform [91]—which stand in contrast to the majority of application papers, especially the recent causal-discovery cluster, that do not emphasize released code, data, or standardized metrics. This gap is partly mitigated by theoretical and complexity-oriented work that provides durable, reproducible guarantees independent of implementation: NP-hardness results for approximate inference [31], polytope and facet analyses for optimal structure learning [17], insensitivity-to-imprecision analyses for diagnostic belief networks [46], and missing-data handling frameworks [2], [3]. Large-scale empirical validations such as [90] and comprehensive surveys [15], [29], [48] play an important role in aggregating and stress-testing algorithmic claims that individual papers rarely benchmark against one another, partially compensating for the field's fragmented evaluation culture.

E. Synthesis

Taken together, the surveyed works depict Bayesian networks as a mature yet actively evolving formalism whose theoretical core—structure learning, exact/approximate inference, and causal semantics—was substantially settled between the 1990s and early 2010s [14], [16], [24]–[26], [31], while its recent trajectory, concentrated in the last three years per Fig. 1, is defined by integration with deep learning, streaming/temporal data, and explainability demands [34], [63], [64], [79], [93]. The dominant open issue is not algorithmic novelty but methodological rigor: the proliferation of similarly framed dynamic/causal Bayesian network variants in 2026 outpaces the field's capacity to benchmark them against shared datasets, common baselines, or openly reproducible implementations, a concern echoed by broader reviews calling for consolidation of causal discovery and probabilistic graphical model research [1], [4], [49].

XII. Open Challenges and Future Directions

A. Scalability and Computational Complexity of Structure Learning

Learning the directed acyclic graph (DAG) that best represents a domain remains one of the most persistent bottlenecks in Bayesian network research. The task is fundamentally a

combinatorial optimization problem: the number of candidate DAGs grows super-exponentially with the number of variables, and the acyclicity constraint prevents the use of standard convex optimization machinery [19], [47], [48]. Surveys covering dozens of structure-learning algorithms confirm that no single method dominates across problem settings, and that determining graph structure under causal assumptions is especially fraught [15]. Integer-programming formulations have improved optimality guarantees, yet foundational understanding of the polytopes underlying these approaches is still incomplete [17], and the sheer size of the search space continues to challenge even hybrid and metaheuristic solvers [20]. Promising directions include tighter integration of constraint propagation with optimization-based search, better theoretical characterization of search-space geometry, and further development of scalable approximate and anytime algorithms that trade optimality for tractability on high-dimensional problems.

B. Tractable Exact and Approximate Inference at Scale

Even when a correct network structure is available, inference—the computation of posterior marginals or conditional probabilities—remains provably hard: approximating conditional probabilities in belief networks is NP-hard, just as exact inference is [31]. Loopy belief propagation and related message-passing schemes offer practical approximations, but their behavior is unpredictable: empirical studies show convergence and good accuracy on some real-world networks while exhibiting oscillation and no clear relationship to correct posteriors on others [26]. Expectation propagation partially generalizes belief propagation to hybrid discrete-continuous settings, yet it remains best suited to restricted classes of networks and lacks guaranteed closed-form solutions in general hybrid models [27], [29]. Applications such as attack-graph analysis further illustrate that exact inference simply does not scale to realistic problem sizes, forcing reliance on approximate methods whose error behavior is not well characterized [94], [96]. Future work should prioritize inference algorithms with provable convergence and error bounds for hybrid and loopy structures, along with principled criteria for predicting when approximate methods will fail before deployment.

C. Causal Discovery from Dynamic, Streaming, and Temporal Data

A large and rapidly growing body of work seeks to extend Bayesian networks beyond static, fully observed snapshots toward settings where data arrives continuously or evolves over time. Traditional causal discovery methods assume access to a complete dataset, a constraint that severely limits applicability in dynamic, evolving environments [36], and conventional algorithms similarly struggle to capture complex temporal dependencies, limiting their interpretability and predictive accuracy [71]. Proposed remedies include dynamic structure-learning algorithms that continuously update topology and parameters as new data arrive [34], [35], Bayesian inference frameworks explicitly designed to mitigate confounding and model temporal dependence [38], integration with Hidden Markov Models to represent latent temporal states [39], [40], and intervention-aware or MCMC-based dynamic Bayesian network formulations tailored to streaming and intervention-laden data [37], [55], [60], [69], [72]. High-dimensional time series pose additional difficulty, as causal relationships must be disentangled from confounding and intricate temporal structure simultaneously [58], and nature-inspired optimization has been proposed to improve dynamic structure search [21]. Because most of these proposals remain theoretical or conceptual frameworks awaiting empirical validation, a key open direction is rigorous benchmarking of dynamic and streaming causal discovery methods against realistic, non-stationary data-generating processes.

D. Latent Confounders, Feedback Loops, and Violations of Causal Sufficiency

Standard causal discovery algorithms typically assume causal sufficiency—that no unmeasured confounder influences multiple observed variables—and acyclicity of the underlying graph, assumptions that are routinely violated in practice. Work on dynamic Bayesian networks explicitly acknowledges that latent variables and feedback loops complicate causal discovery and require dedicated modeling machinery rather than being treated as noise [39]. Confounding factors intertwined with temporal dependencies are similarly flagged as a core obstacle to reliable causal inference from time-varying data [38]. More broadly, large-scale empirical validation shows that structure-learning algorithms are typically developed and evaluated under assumptions of complete data and causal sufficiency that are unrealistic in real-world deployments, meaning synthetic benchmark performance likely overestimates real-world performance to an unknown degree [90]. Addressing this gap calls for structure-learning methods that explicitly model latent confounding (e.g., via latent-variable DAGs or ancestral graphs), feedback-aware extensions of dynamic Bayesian networks, and evaluation protocols using data that deliberately violate idealized assumptions so that robustness, not just accuracy under ideal conditions, can be measured.

E. Robustness to Missing, Noisy, and Non-Ideal Data

Real-world datasets are rarely complete, yet most structure-learning theory is developed under an implicit assumption of fully observed data. Tools such as bnstruct were developed specifically because, to the best of the authors' knowledge, no other open-source software jointly supported structure learning, parameter learning, and inference in the presence of missing values [3], underscoring how underserved this problem has been at the tooling level. On the theoretical side, graphical-model-based procedures have been shown to overcome limitations of conventional missing-data methods and to provide performance guarantees even under Missing Not At Random (MNAR) mechanisms [2], yet such guarantees are not uniformly available across algorithm families. Compounding this, empirical validation under noisy, real-world-like data reveals that algorithm rankings and performance can diverge substantially from idealized benchmarks [90]. Future research should focus on extending missing-data-aware inference and learning theory to a broader range of

structure-learning algorithms, developing standardized noisy-data benchmarks, and building accessible open-source tooling that makes robust handling of imperfect data the default rather than the exception.

F. Hybrid Neuro-Symbolic Integration with Deep Learning

A recurring theme across recent proposals is combining the pattern-recognition strength of neural networks with the structured, logical inference capabilities of Bayesian networks, precisely because neither paradigm alone suffices for complex, high-dimensional, and partially interpretable reasoning tasks. Frameworks merging neural pattern recognition with dynamic Bayesian network inference are explicitly motivated by the limitations of purely data-driven or purely rule-based diagnostic methods [63]. Similarly, graph-based reasoning combined with Bayesian network principles has been proposed to iteratively refine causal graph structure while enforcing consistency constraints, with explicit calls for future work combining such hybrids with reinforcement learning and domain knowledge [66]. Deep neural networks have also been proposed to directly learn conditional independence structure from data, moving beyond strong parametric assumptions or reliance on expert knowledge [65], while symbolic reasoning engines have been integrated with data-driven causal discovery specifically to curb spurious correlations and restore interpretability [64]. Generative deep learning has further been proposed as a density-free route to Bayesian posterior computation [45], and Bayesian approximations such as DropConnect illustrate the broader push to equip deep networks with calibrated

XIII. Conclusion

Bayesian networks have evolved from a formalism for compact probabilistic representation into a central methodological bridge connecting statistical inference, causal reasoning, and modern AI systems. The taxonomy developed in this survey organizes the literature into seven interrelated strands: causal discovery from observational and streaming data, the integration of probabilistic graphical models within broader AI architectures, causal Bayesian networks for medical decision-making, the foundational problems of structure learning and conditional independence testing, exact and approximate inference algorithms, metaheuristic approaches to structure optimization, and the computational complexity results that bound what is achievable in all of the above. Across these strands, a consistent pattern emerges: the field has progressively moved from static, hand-specified networks toward dynamic, data-driven, and hybrid neuro-probabilistic models, exemplified by dynamic Bayesian network formulations for streaming and time-series causal discovery [34]–[36], [38] and by architectures that fuse graphical models with neural components for scalable inference and representation learning [28], [43], [45]. At the same time, classical contributions on structure learning algorithms, conditional independence foundations, and exact/approximate inference [14], [15], [24]–[27], [47] remain the backbone upon which these newer developments are built, while metaheuristic optimization techniques such as genetic algorithms, pigeon-inspired optimization, and hybrid swarm methods [18]–[20], [22], [100] offer practical means of navigating the combinatorially vast space of candidate network structures.

A central finding is that the appeal of Bayesian networks lies in their dual capacity to support both predictive inference and causal explanation, a property that has made them especially prominent in high-stakes, interpretability-sensitive domains such as medical diagnosis, where causal Bayesian networks are used to model symptom-disease-treatment pathways and to generate human-interpretable explanations of AI predictions [7], [12], [13], [80], [86]. This interpretability, however, is counterbalanced by fundamental computational limits: exact inference and even approximate inference in general Bayesian networks are provably intractable in the worst case [31], and optimal structure learning is similarly hard, as formalized through dynamic programming and integer programming analyses of the search space [16], [17]. Reviews and surveys of structure learning and of the broader probabilistic graphical model landscape [4], [15], [48], [49] confirm that no single algorithmic paradigm dominates; rather, practitioners must trade off accuracy, scalability, and robustness to missing or noisy data [2], [3] depending on application constraints, as also illustrated by decision-theoretic and autonomous-systems use cases [10], [41], [78].

Looking forward, the most pressing directions concern reconciling the scalability demands of modern AI with the interpretability and causal rig

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